

Winning Space Race with Data Science

Sizwe Shongwe
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Space X Data Collection using Space X API
- Space X Data Collection with Web Scraping
- Space X Data Wrangling
- Space X Exploratory Data Analysis using SQL
- Space X EDA DataViz Using Python Pandas and Matplotlib
- Space X Launch Sites Analysis with Folium Interactive Visual Analytics and PlotyDash
- Space X Machine Learning Landing Prediction

Summary of all results:

- EDA results
- Interactive Visual Analytics and Dashboards
- Predictive Analysis(Classification)

Introduction

Project Background and Context

- Private and commercial space launches have become increasingly competitive because reusable rockets can reduce launch costs and make space missions more economically sustainable. In this project, I analyse SpaceX launch data to understand the factors associated with successful launches and successful booster landings.
- The project uses data collected from the SpaceX REST API and web scraping. The dataset was cleaned and prepared using SQL, Python, and pandas before being explored through statistical analysis, charts, interactive dashboards, and geospatial maps.
- Several launch and operational factors were studied, including launch site, flight number, payload mass, orbit type, booster version, launch year, and landing outcome. Finally, classification models were built to predict whether a rocket landing would be successful based on available launch features.

Problems to Find Answers

- This project aims to answer the following questions:
- How many successful and unsuccessful launches occurred, and how do outcomes differ between launch sites?
- Which launch sites are associated with the highest launch success rates?
- Does flight experience affect launch success?
- How does payload mass influence launch success?
- Which orbit types have the highest success rates?
- What has the yearly trend in launch success been?
- Where are the SpaceX launch sites located?
- What are the distances between selected launch sites and nearby railway lines, highways, coastlines, and other infrastructure?
- Which factors help determine whether a booster landing is successful?
- Can machine-learning models accurately predict launch landing success?
- Which classification model produces the best predictive performance?

Section 1

Methodology



Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

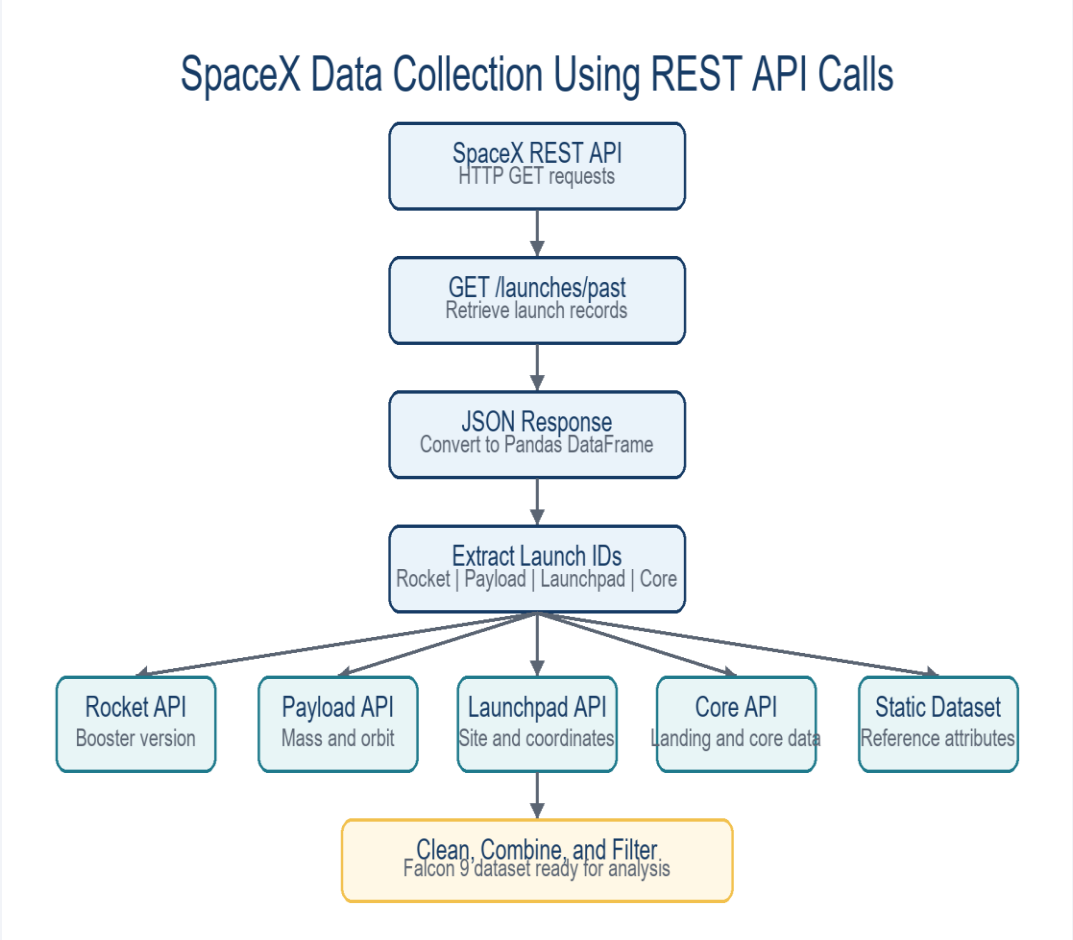
Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

- Collected historical SpaceX launch data through REST API requests.
- Used HTTP GET requests to retrieve launch information.
- Extracted nested JSON fields from multiple SpaceX API endpoints.
- Converted API responses into Pandas DataFrames.
- Used API identifiers to retrieve related rocket, payload, launch-site, and core details.
- Filtered the data to Falcon 9 launches.
- Produced a structured dataset for exploratory analysis and predictive modelling.

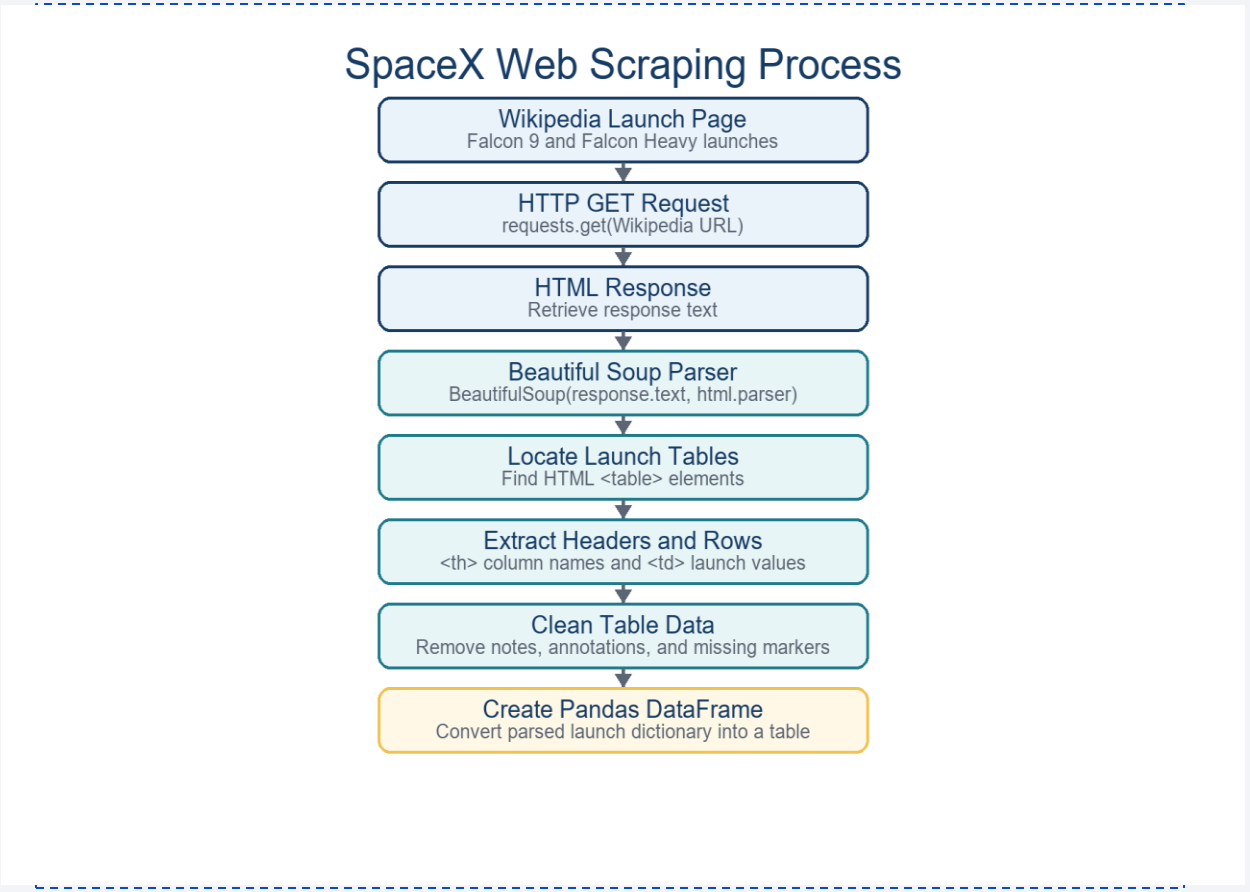
URL: [Github Data Collection Link]



Data Collection - Scraping

- Send an HTTP request to the Wikipedia page.
- Retrieve the launch records as HTML content.
- Parse HTML using Beautiful Soup.
- Locate Falcon 9 and Falcon Heavy launch tables.
- Extract table headers using `<th>` elements.
- Extract launch values using `<td>` elements.
- Remove reference notes, annotations, and missing markers.
- Store extracted records in a launch dictionary.
- Convert the dictionary into a Pandas DataFrame.
- Create a structured dataset for further analysis.

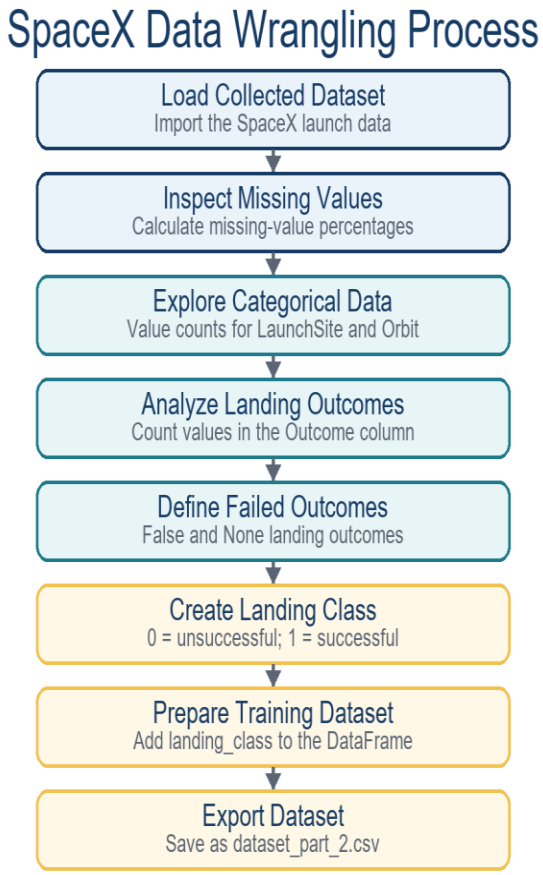
URL:[Github Web Scraping Link]



Data Wrangling

- Load the collected SpaceX dataset.
- Inspect missing-value percentages.
- Analyse LaunchSite, Orbit, and Outcome.
- Apply frequency counts using value_counts().
- Identify successful and unsuccessful landing outcomes.
- Define failed outcomes as 0.
- Define successful outcomes as 1.
- Create the landing_class training label.
- Export the processed data as dataset_part_2.csv.

URL:Github Data Wrangling Link



EDA with Data Visualization

- **Scatter-style categorical plot: Payload Mass vs. Flight Number**
  - Used a `catplot` with landing outcome represented by colour.
  - Showed whether payload mass and mission sequence influenced launch success or failure.
- **Categorical plot: Flight Number vs. Launch Site**
  - Compared SpaceX flights across the different launch sites.
  - Used colour to distinguish successful and failed landings.
- **Categorical plot: Payload Mass vs. Launch Site**
  - Examined whether payload mass differed between launch sites.
  - Compared landing outcomes for each launch-site and payload combination.
- **Bar chart: Success Rate by Orbit Type**
  - Plotted the mean binary landing-class value for each orbit.
  - Compared the relative landing success rates across orbit types.
- **Categorical plot: Flight Number vs. Orbit**
  - Visualized the orbit used by each flight over time.
  - Used landing outcome as the colour distinction to identify changes in success across mission types.
- **Categorical plot: Payload Mass vs. Orbit**
  - Examined the relationship between payload mass and orbit type.
  - Used colour to compare successful and failed landings for similar payload ranges.
- **Line chart: Annual Success Rate**
  - Grouped landing outcomes by year and plotted their mean success rate.
  - Used a line chart to show trends and changes in SpaceX landing performance over time.

URL:GIthub Data Visualization Link

EDA with SQL

- Created an SQLite database connection and loaded the SpaceX dataset.
- Removed the existing table and recreated the cleaned SPACEXTABLE.
- Displayed unique launch sites using SELECT DISTINCT.
- Retrieved five SpaceX records where launch sites begin with CCA.
- Calculated total payload mass for NASA CRS missions using SUM().
- Calculated average payload mass for Falcon 9 version F9 v1.1 using AVG().
- Identified the earliest successful ground-pad landing using MIN().
- Filtered boosters that successfully landed on a drone ship and carried payloads between 4,000 and 6,000 kg.
- Counted successful and failed mission outcomes.
- Used a subquery to identify booster versions carrying the maximum payload mass.
- Extracted 2015 drone-ship landing failures by month, booster version, and launch site.
- Grouped and counted landing outcomes by type within the specified date range and ranked them in descending order.

URL:Github SQL Link

Build an Interactive Map with Folium

- **Base map**
  - Created a Folium map centred near NASA's Johnson Space Center.
  - Used the map to visually compare launch sites, mission outcomes, and nearby infrastructure.
- **Launch-site circles**
  - Added a folium.Circle for each launch site with a radius of 1 km.
  - Used these circles to highlight and distinguish the exact location of each launch facility.
  - Added a popup containing the launch-site name.
- **Launch-site labels**
  - Added DivIcon markers displaying each launch-site name.
  - These labels made it easier to identify launch sites directly on the map.
- **Launch outcome markers**
  - Added launch-coordinate markers to a `MarkerCluster`.
  - Used green markers for successful landings and red markers for failed landings.
  - Cluster markers helped manage many mission locations while preserving outcome colour coding.
- **Proximity markers**
  - Added a distance label marker for the closest coastline point.
  - Added distance-marker objects for a nearby city, railway, and highway.
  - Used `DivIcon` text markers to display calculated distances in kilometres.
- **Distance lines**
  - Added `PolyLine` objects connecting each selected proximity to the launch site.
  - The lines visually showed the direction and separation between the launch facility and coastline, city, railway, and highway.

Build an Interactive Map with Folium

- **Purpose**
  - To identify the geographic locations of SpaceX launch facilities.
  - To examine the spatial distribution of successful and failed launches.
  - To evaluate how launch sites are positioned relative to cities, transportation routes, and coastlines.
  - To determine whether launch facilities maintain reasonable distances from populated or infrastructure-sensitive areas.
  - To support geographic risk analysis and launch-site suitability assessments.

URL:Github Mapping with Folium

Build a Dashboard with Plotly Dash

- **Launch-site dropdown**
  - Allowed users to select an individual launch site or all launch sites.
  - Made the dashboard interactive and enabled site-specific analysis.
- **Success pie chart**
  - For all sites, displayed the proportion of successful launches contributed by each launch site.
  - For a selected site, displayed the proportion of successful and failed missions.
  - Used a pie chart to clearly compare categorical launch outcomes and contributions.
- **Payload range slider**
  - Allowed users to select a minimum and maximum payload mass.
  - Filtered mission records to a specified payload range.
- **Payload versus launch outcome scatter plot**
  - Plotted payload mass against mission outcome, with booster category shown by colour.
  - Allowed users to examine whether heavier or lighter payloads were associated with successful launches.
  - Updated according to the selected launch site and payload range.

Build a Dashboard with Plotly Dash

- **Purpose of the Interactions**
  - **Dropdown and range slider callbacks** refreshed the charts automatically.
  - **Site filtering** helped identify launch-site-specific performance.
  - **Payload filtering** allowed comparison of outcomes across payload categories.
  - **Combined controls** supported analysis of the relationship between payload mass, booster version, launch site, and landing success.

URL:Github Building dashboard with Plotly

Predictive Analysis (Classification)

- **Input Data:** SpaceX mission and landing features
- **Preprocessing:** Feature encoding and StandardScaler normalization
- **Train/Test Split:** 80% training and 20% testing
- **Test Dataset:** 18 observations, with random_state=2
- **Classification Models:**
  - Logistic Regression
  - Support Vector Machine
  - Decision Tree
  - K-Nearest Neighbors
- **Hyperparameter Tuning:** Grid search with 10-fold cross-validation
- **Validation Accuracy:**
  - Logistic Regression: 84.64%
  - SVM: 84.82%
  - Decision Tree: 86.07%
  - KNN: 84.82%
- **Model Evaluation:** Accuracy scores and confusion matrices
- **Test Accuracy:** All four models achieved 83.33%
- **Best Model:** Decision Tree achieved the highest cross-validation accuracy at 86.07%

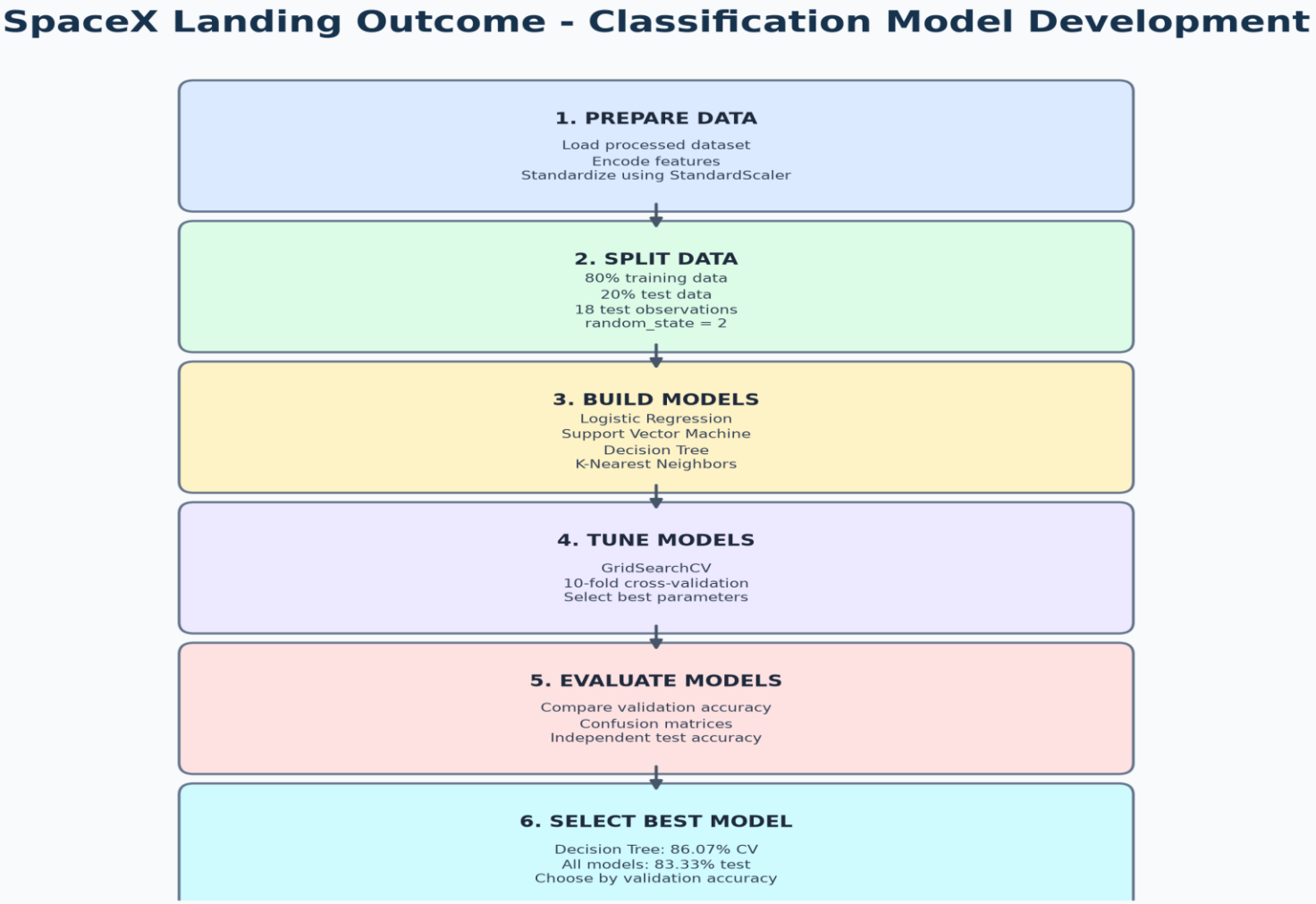
Predictive Analysis (Classification)

- Loaded the processed launch dataset and separated features from the landing outcome.
- Encoded categorical features and standardized the feature values.
- Split the data into 80% training and 20% test sets.
- Built four classification models to predict successful or unsuccessful landings.
- Used `GridSearchCV` with 10-fold cross-validation to select the best hyperparameters.
- Compared model validation performance using accuracy.
- Evaluated predictions using the independent test dataset and confusion matrices.
- Selected the Decision Tree model based on its highest cross-validation accuracy of 86.07%.

All models achieved the same 83.33% accuracy on the test set, while the Decision Tree performed best during hyperparameter tuning.

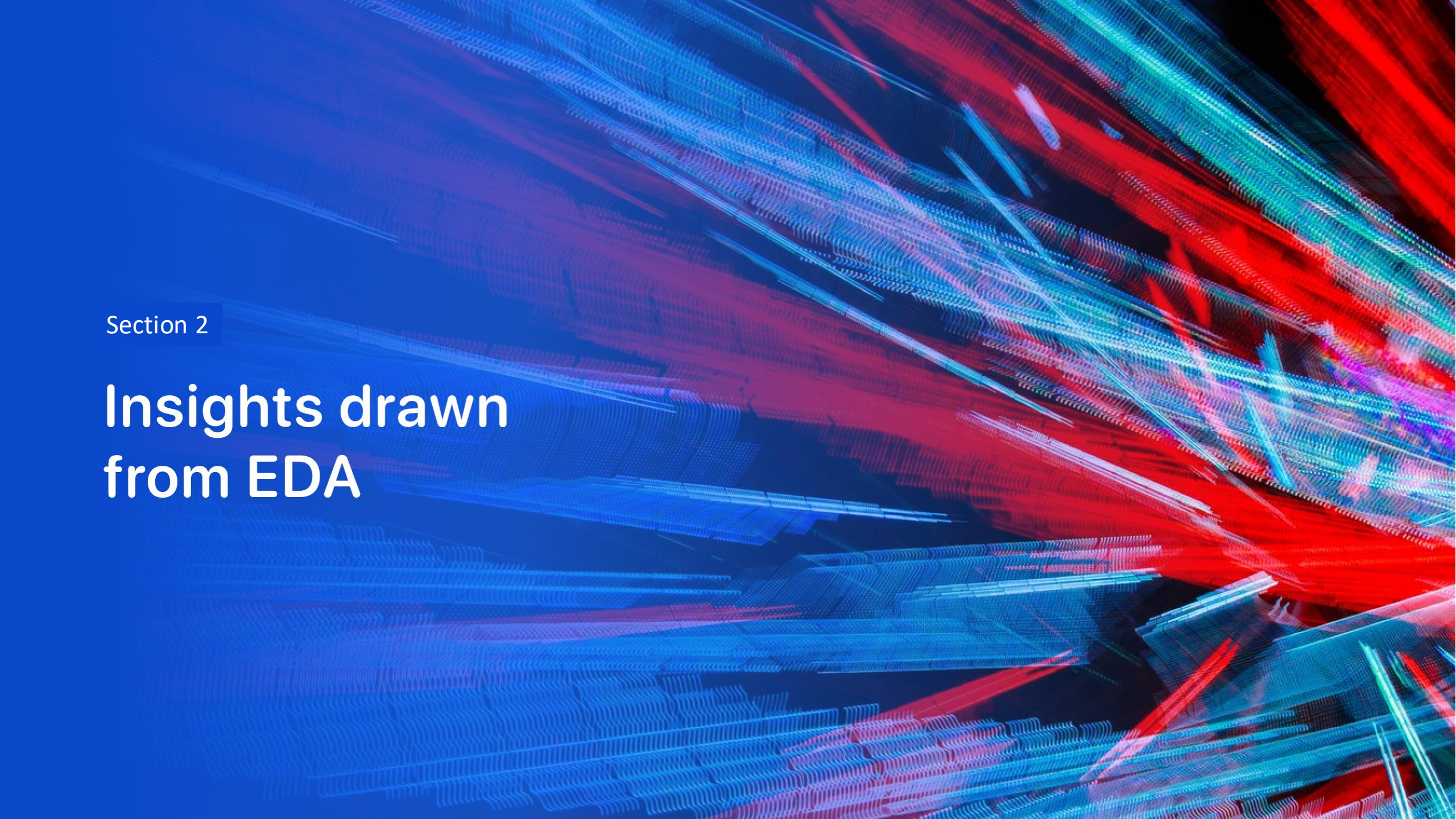
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Predictive Analysis (Classification)



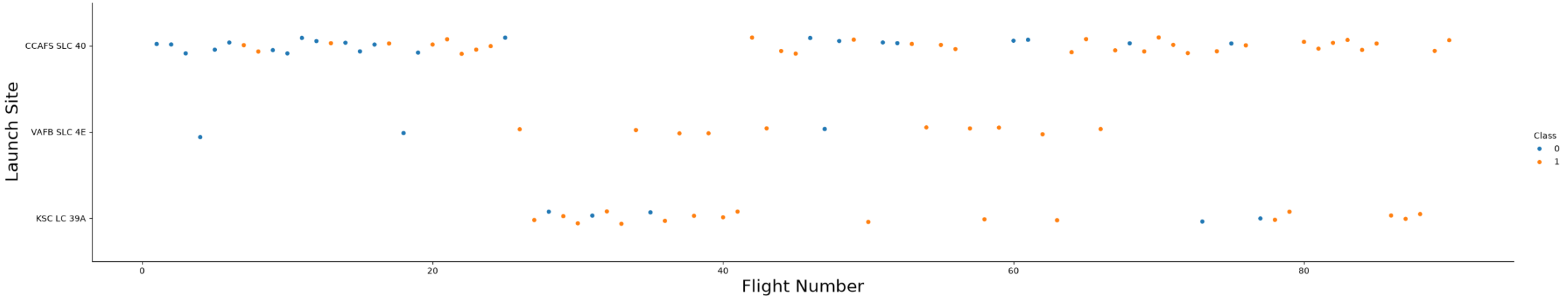
Results

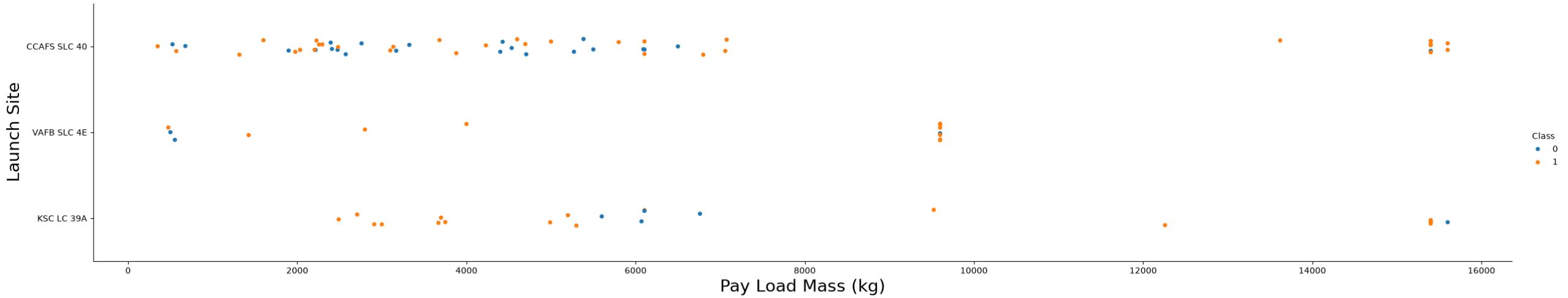
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results


Section 2
Insights drawn from EDA

Flight Number vs. Launch Site

This scatter plot shows the number of flights launched from each launch site over time. Each point represents a launch, with colour indicating whether the landing was successful or failed. The plot shows that launch activity increased across later flight numbers, particularly at certain sites, and it also highlights differences in landing success between launch locations.



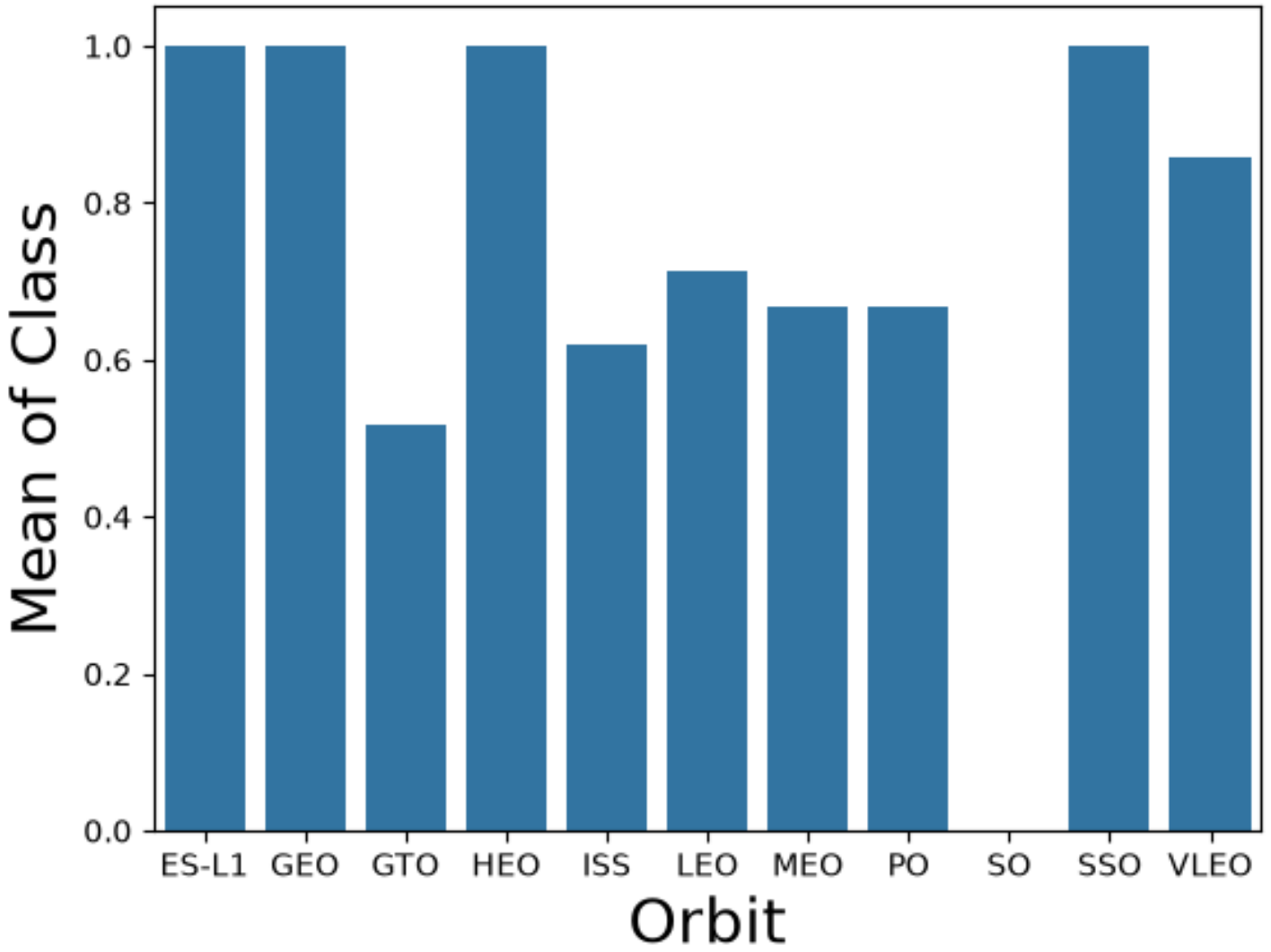


Payload vs. Launch Site

This scatter plot compares payload mass with launch-site location. Each point represents one mission, coloured by landing outcome. It shows that launch sites handled different payload ranges and helps identify whether payload weight was associated with landing success at a particular site.

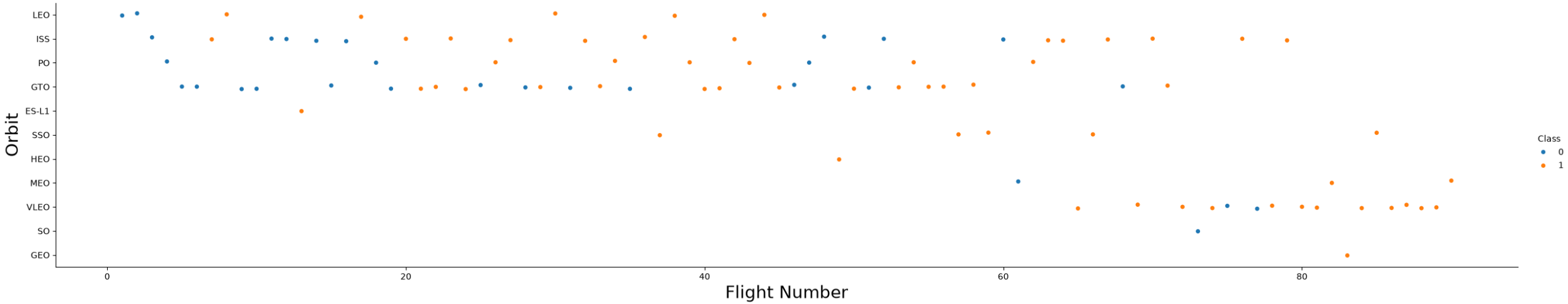
Success Rate vs. Orbit Type

The bar chart displays the average landing outcome for each orbit type. Taller bars indicate a higher success rate, while shorter bars indicate lower success. It helps compare the reliability of launches across orbit types and identify orbits associated with stronger or weaker landing performance.



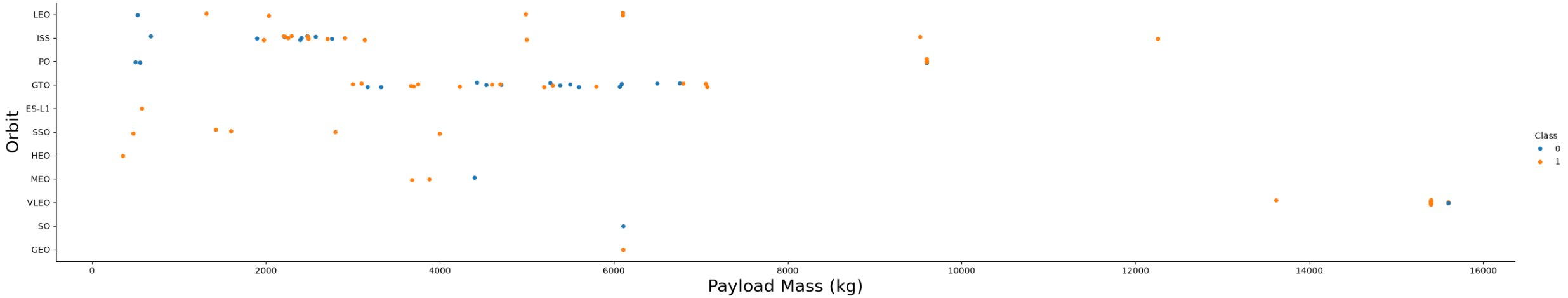
Flight Number vs. Orbit Type

This scatter plot shows how the selected orbit types changed as launch numbers increased. Each point represents a mission, with colour indicating landing success or failure. It reveals whether SpaceX expanded into particular orbit types over time and whether mission outcomes differed by orbit and flight progression.



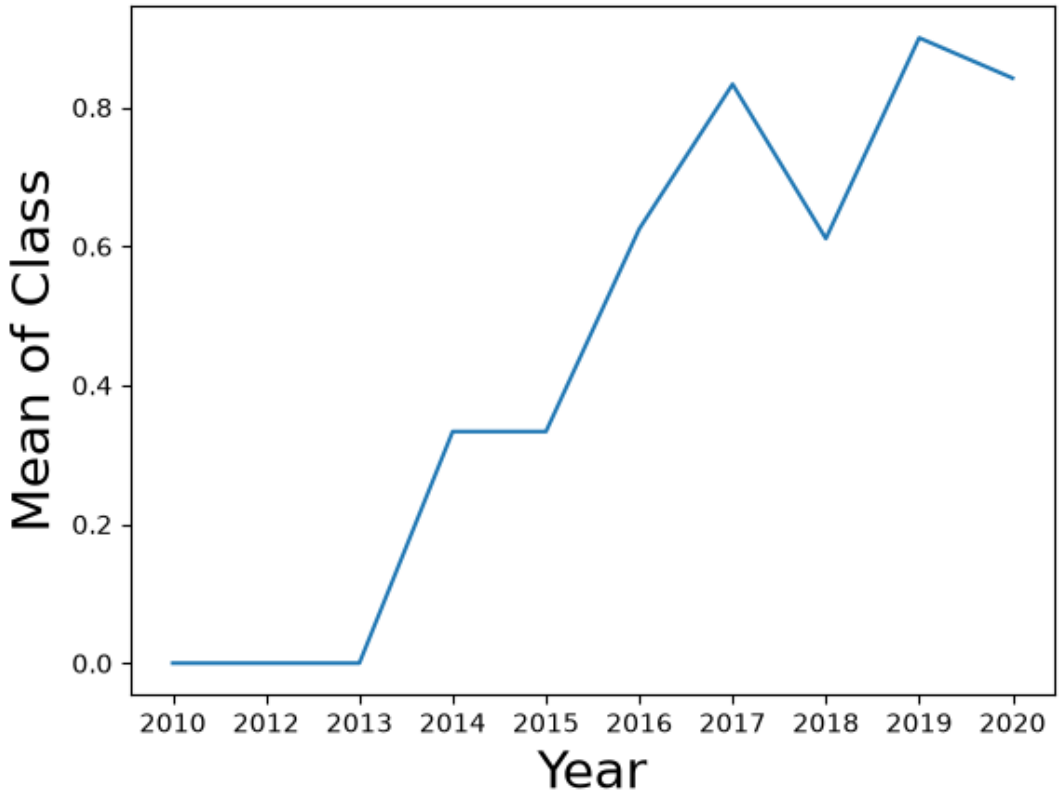
Payload vs. Orbit Type

This scatter plot compares payload mass across orbit types and distinguishes successful and unsuccessful landings by colour. It helps identify which orbits carried heavier payloads and whether payload weight was associated with landing performance for particular orbit types.



Launch Success Yearly Trend

The line chart shows the mean landing outcome for each year. An upward trend indicates improving landing success over time, while downward movements indicate periods with lower success rates. This visualization helps summarize SpaceX's improvement in landing performance and highlights changes in success over its launch history.



All Launch Site Names

These are the four unique launch-site locations recorded in the dataset.

Task 1

Display the names of the unique launch sites in the space mission

```
[13]  1  %sql select distinct "Launch_Site" from spacextable
```

Python

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Five launch records beginning with CCA, including CCAFS LC-40 missions from 2010, 2012, and 2013.

Task 2

Display 5 records where launch sites begin with the string 'CCA'

Generate | + Code | + Markdown

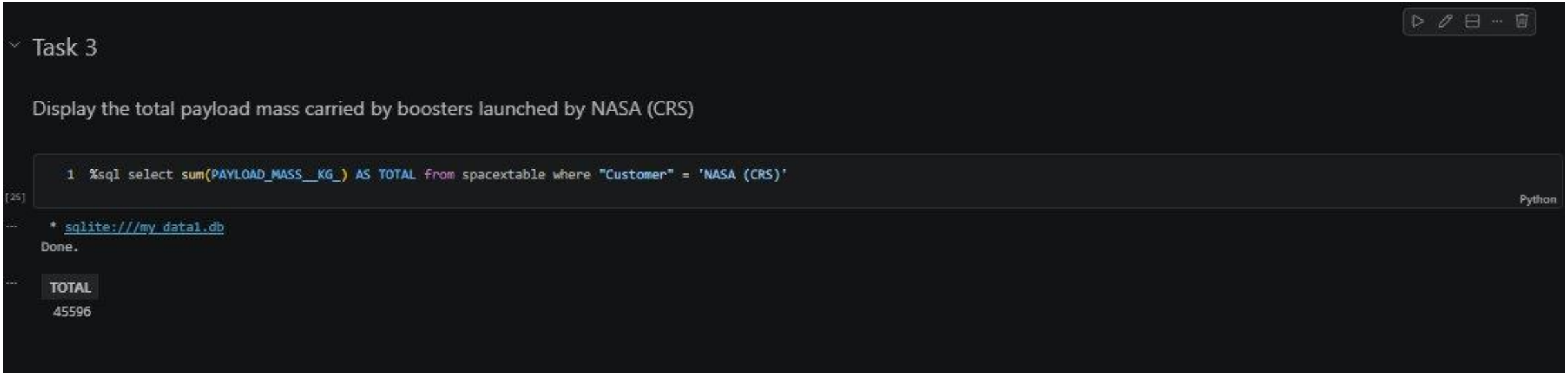
```
[17]  1  %sql select * from spacextable where "Launch_Site" like 'CCA%' limit 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

This is the combined payload mass of all missions in the dataset where the customer is NASA CRS.



Task 4

Display average payload mass carried by booster version F9 v1.1

```
[26]  1  %sql select avg(PAYLOAD_MASS__KG_) AS AVG from spacextable where "Booster_Version" = 'F9 v1.1'
                                                                                                  Python
```

```
 * sqlite:///my_data1.db
Done.
```

AVG
2928.4

Average Payload Mass by F9 v1.1

- This is the mean payload mass carried by boosters identified as F9 v1.1.

First Successful Ground Landing Date

This is the earliest recorded successful landing on a ground pad.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
[27]  1  %sql select min("Date") from spacextable where "Landing_Outcome" = 'Success (ground pad)'
                                                                                              Python
```

```
 * sqlite:///my_data1.db
Done.
```

min(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- These booster versions successfully landed on a drone ship while carrying payloads in the specified range.

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[29]  1  %sql select "Booster_Version" from spacextable where "Landing_Outcome" = 'Success (drone ship)' and payload_mass__kg_ between 4000 and 6000
                                                                                                                                    Python
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The result represents the total number of records classified as either a successful or failed mission outcome.

Task 7

List the total number of successful and failure mission outcomes

```
[31]  1  %sql select count(*) from spacextable where "Mission_Outcome" = 'Success' or "Mission_Outcome" = 'Failure'
                                                                                                          Python
...   * sqlite:///my_data1.db
Done.

...   count(*)
      98
```

Boosters Carried Maximum Payload

These boosters share the maximum payload mass recorded in the dataset.

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql select "Booster_Version" from spacextable where payload_mass__kg_ = (select max(payload_mass__kg_) from spacextable)
```

Python

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Two drone-ship landing failures occurred in 2015: one in January and one in April, both launched from CCAFS LC-40.

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[49]
1 %sql select substr(Date, 6, 2) as Month, "Landing_Outcome", "Booster_Version", "Launch_Site" from spacextable where "Landing_Outcome" = 'Failure (drone ship)' and substr(Date, 0, 5) = '2015'
```

Python

```
 * sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- No landing attempts were the most frequent outcome in the notebook's selected date range. The notebook query uses **2017-03-30**, while the task text specifies **2017-03-20**.

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

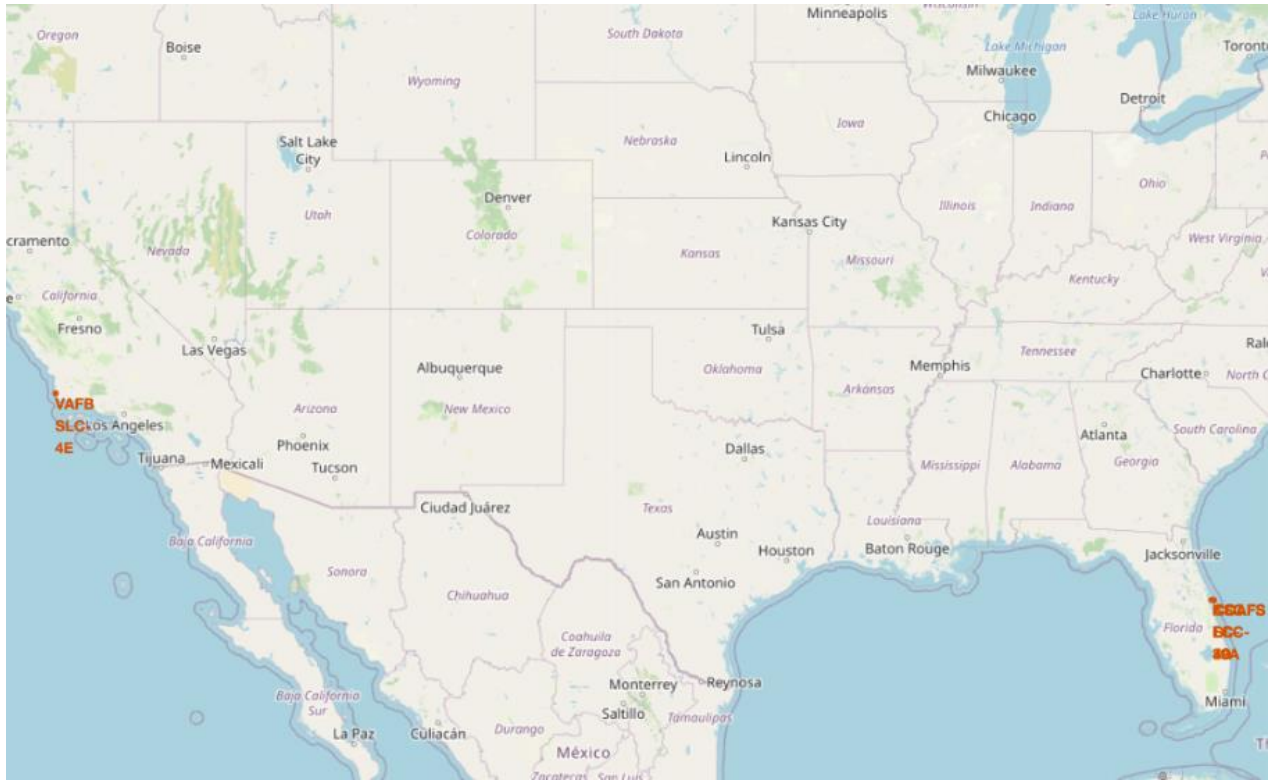
```
[51]  1  %sql select count(*), "Landing_Outcome" from spacextable where Date between '2010-06-04' and '2017-03-30' group by "Landing_Outcome" order by count(*) desc
                                                                                                                                                                    Python
```

```
 * sqlite:///my_data1.db
Done.
```

count(*)	Landing_Outcome
10	No attempt
6	Success (drone ship)
5	Failure (drone ship)
3	Success (ground pad)
3	Controlled (ocean)
2	Uncontrolled (ocean)
2	Failure (parachute)
1	Precluded (drone ship)


Section 3
Launch Sites
Proximities Analysis

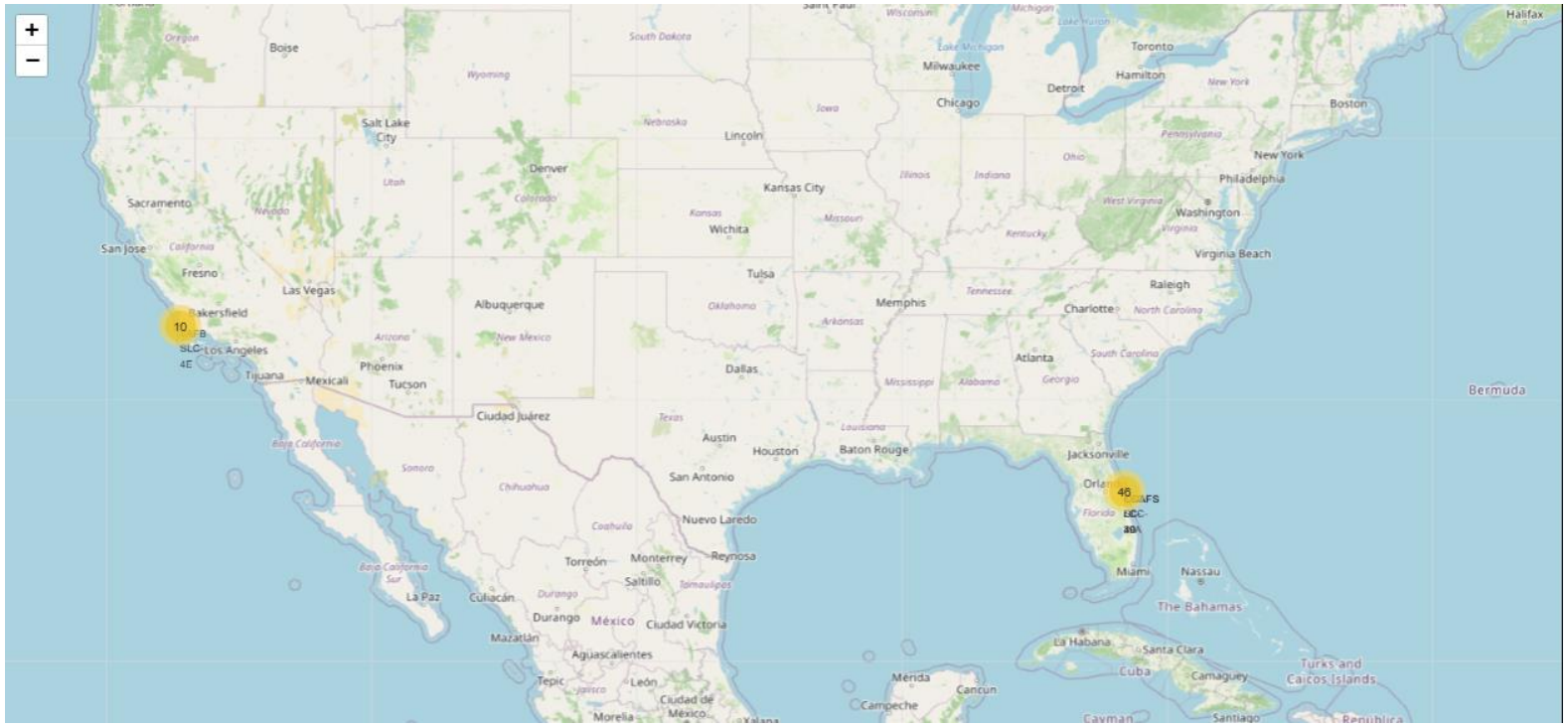
Global View of SpaceX Launch Site



- The pie chart compares successful launches across the four SpaceX launch sites.
- KSC LC-39A generally contributes the largest share of successful launches.
- CCAFS LC-40, CCAFS SLC-40, and VAFB SLC-4E represent the remaining successful-launch distribution.
- Each pie slice shows the proportion of successful launches associated with a site.
- This view helps identify the launch sites contributing most strongly to successful mission totals.

Launch Outcomes by Location

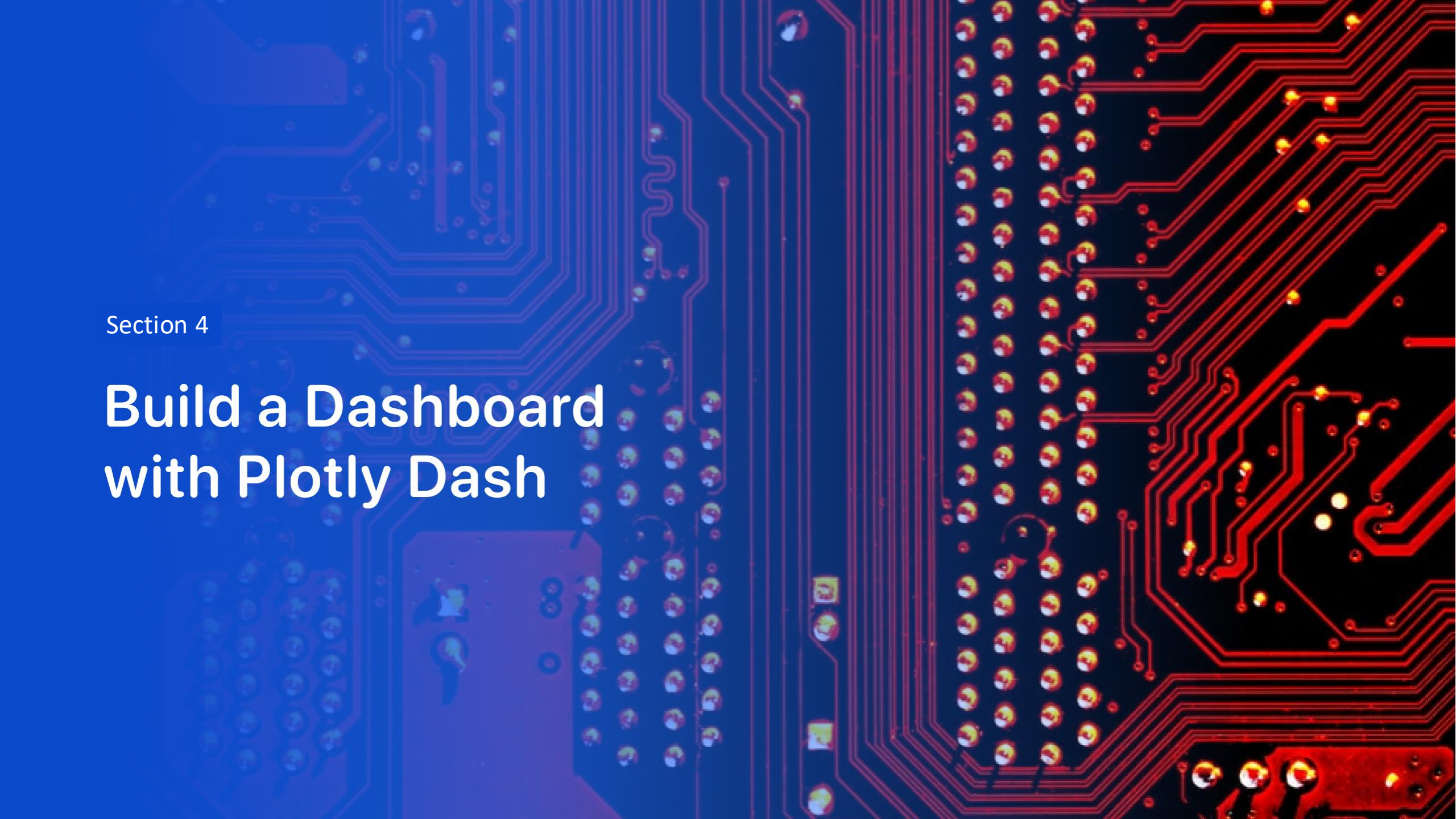
- KSC LC-39A is expected to have the highest successful-launch percentage.
- The pie chart separates successful and failed launches for the selected site.
- The larger pie segment represents successful landings.
- The smaller segment represents unsuccessful or failed landings.
- Site selection is applied through the dashboard launch-site dropdown.
- This view is useful for comparing individual-site reliability rather than overall launch volume.





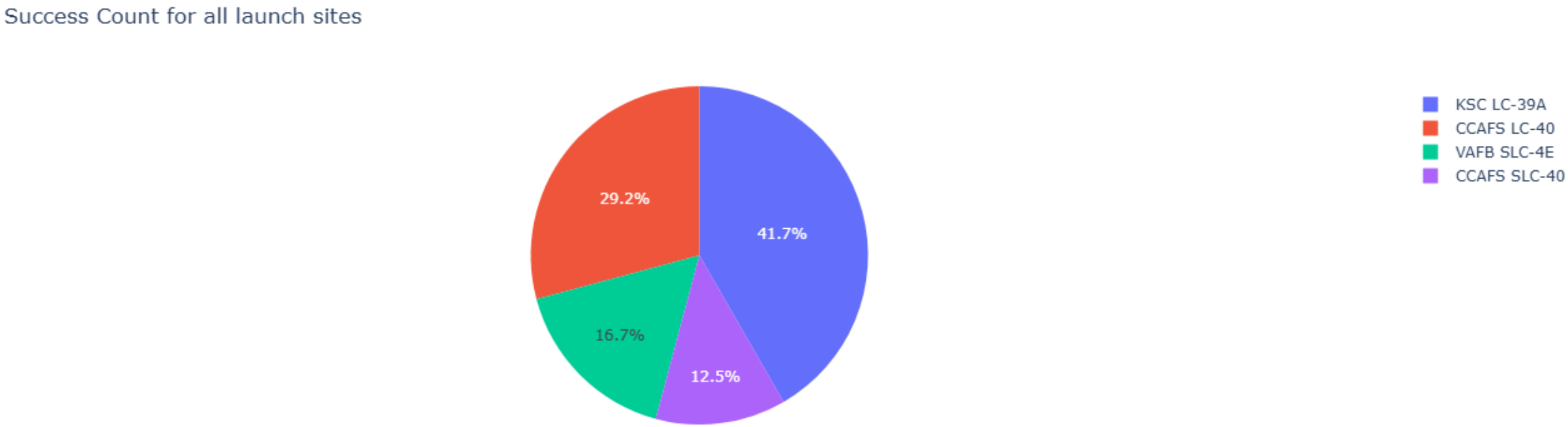
Launch-Site Proximity Analysis

- The x-axis represents payload mass in kilograms.
- The y-axis represents the launch outcome, where 1 indicates success and 0 indicates failure.
- Different colours represent different booster-version categories.
- Increasing the payload range on the slider filters the displayed missions.
- Lower payloads contain both successful and unsuccessful outcomes, while heavier payload missions are generally associated with more recent booster versions.
- The Falcon 9 Block 5 category typically shows the highest concentration of successful launches, especially in higher payload ranges.
- Block 5 booster missions demonstrate strong success performance across medium and high payload ranges.
- No single payload range guarantees success; outcome patterns should be interpreted together with booster category and launch-site performance.
- Adjusting the slider allows comparison of success patterns between lower payloads, medium payloads, and payloads approaching the maximum range.


Section 4
Build a Dashboard with Plotly Dash

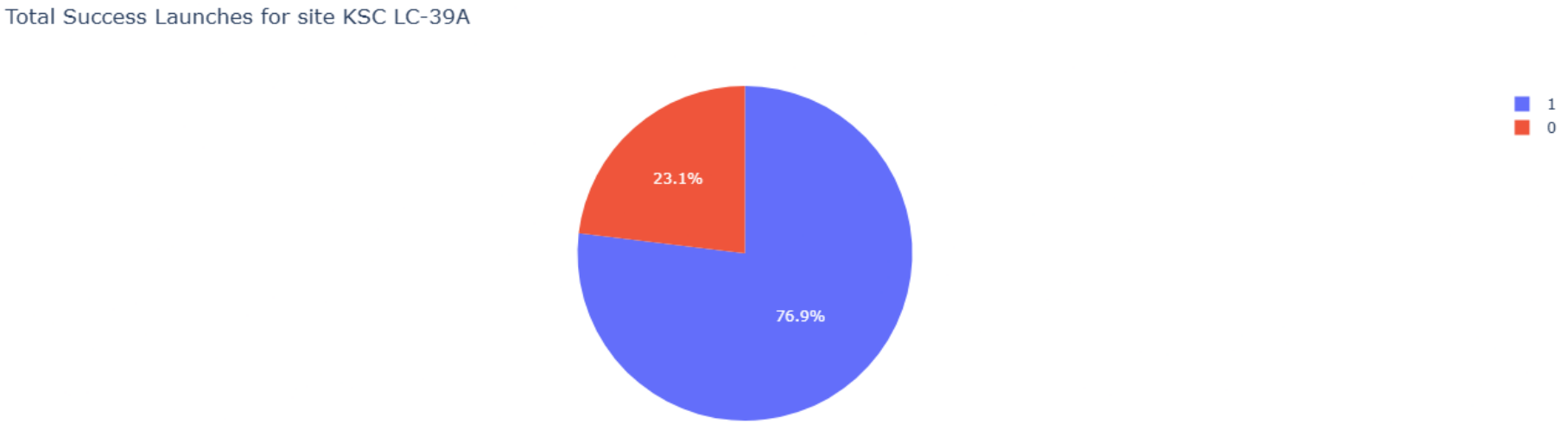
Successful Launches by SpaceX Launch Site

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- CCAFS LC-40, CCAFS SLC-40, and VAFB SLC-4E represent the remaining successful-launch distribution.
- Each pie slice shows the proportion of successful launches associated with a site.
- This view helps identify the launch sites contributing most strongly to successful mission totals.



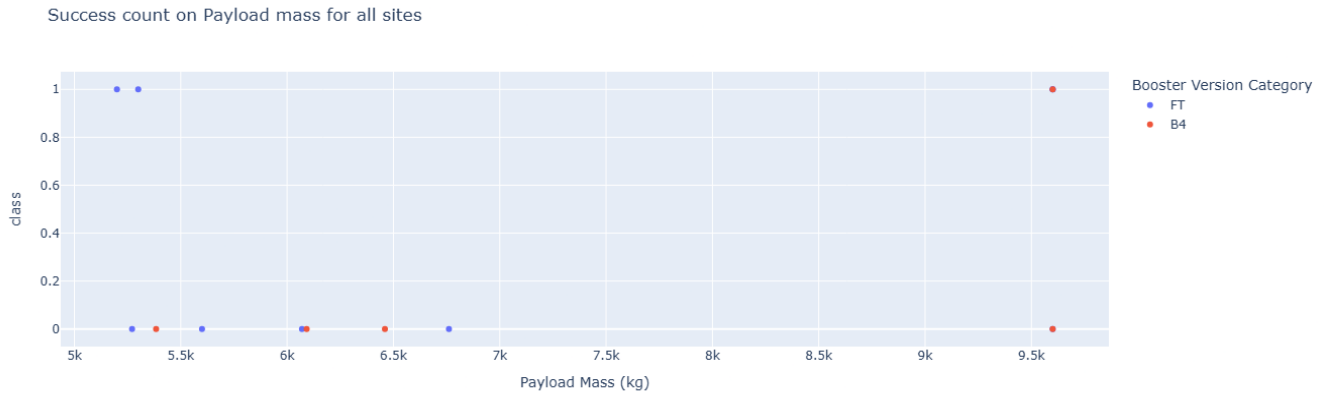
Successful Launches by SpaceX Launch Site

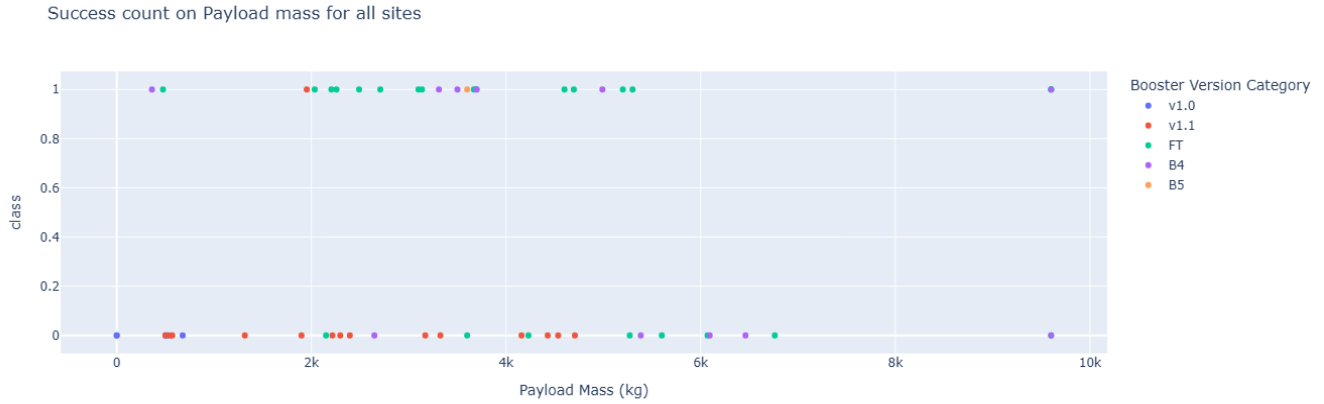
- KSC LC-39A is expected to have the highest successful-launch percentage.
- The pie chart separates successful and failed launches for the selected site.
- The larger pie segment represents successful landings.
- The smaller segment represents unsuccessful or failed landings.
- Site selection is applied through the dashboard launch-site dropdown.
- This view is useful for comparing individual-site reliability rather than overall launch volume.



Payload Mass and Landing Outcome by Booster Version

- The x-axis represents payload mass in kilograms.
- The y-axis represents the launch outcome, where 1 indicates success and 0 indicates failure.
- Different colours represent different booster-version categories.
- Increasing the payload range on the slider filters the displayed missions.
- Lower payloads contain both successful and unsuccessful outcomes, while heavier payload missions are generally associated with more recent booster versions.
- The Falcon 9 Block 5 category typically shows the highest concentration of successful launches, especially in higher payload ranges.
- Block 5 booster missions demonstrate strong success performance across medium and high payload ranges.
- No single payload range guarantees success; outcome patterns should be interpreted together with booster category and launch-site performance.
- Adjusting the slider allows comparison of success patterns between lower payloads, medium payloads, and payloads approaching the maximum range.

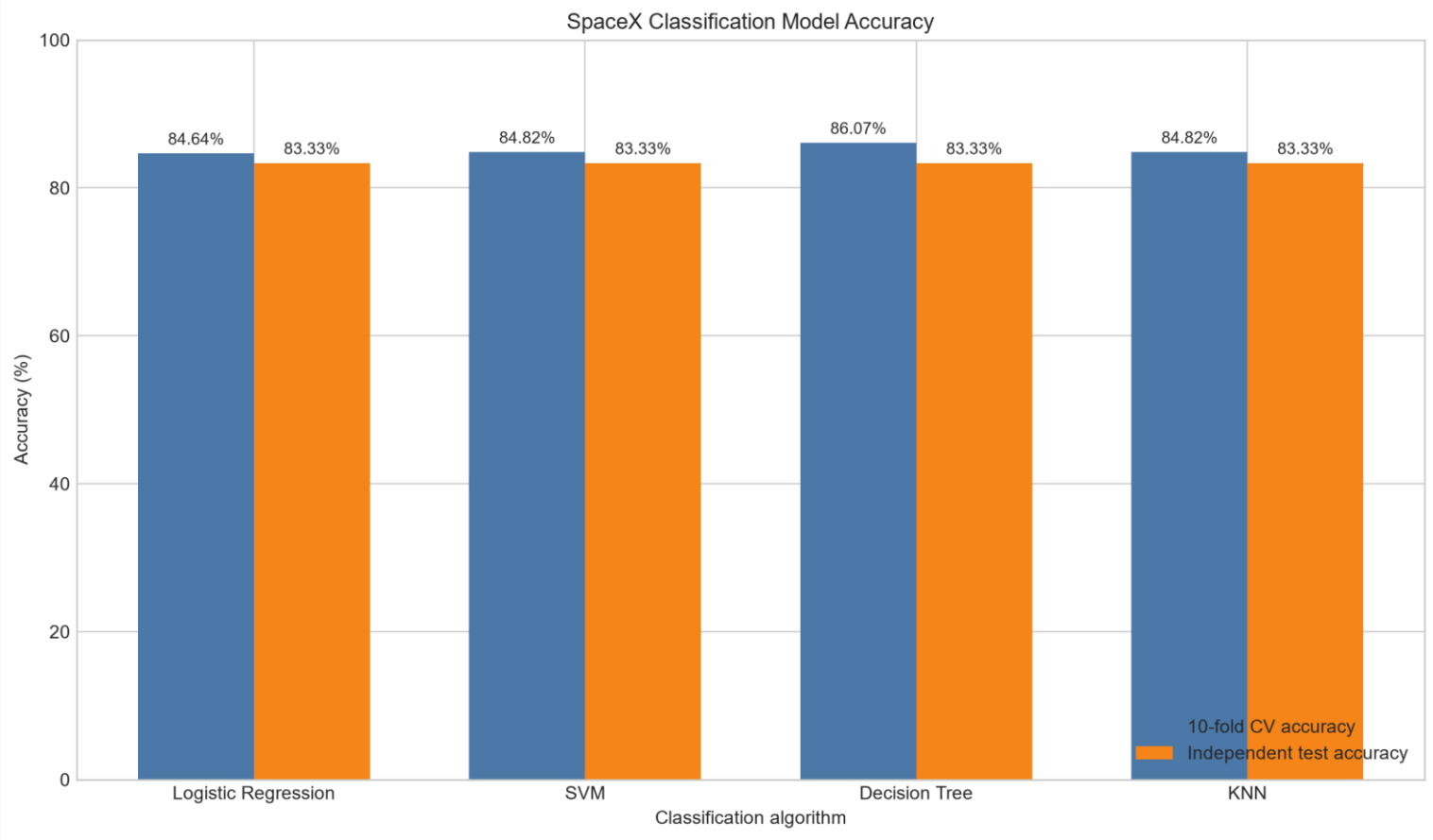




Section 5

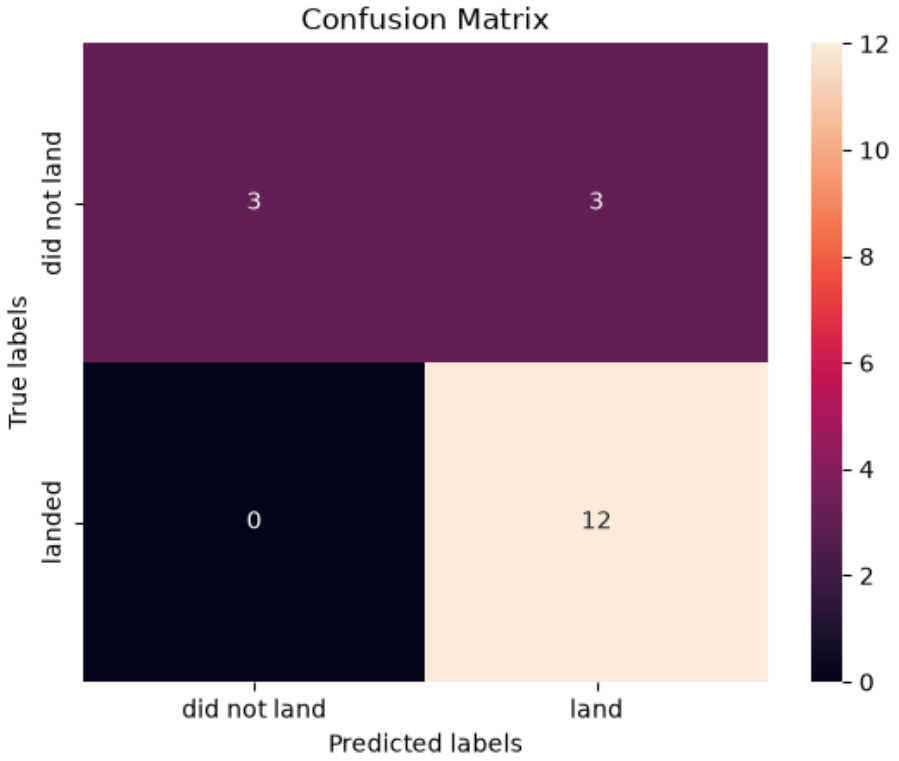
Predictive Analysis (Classification)





Classification Accuracy

- The **Decision Tree** achieved the highest **10-fold cross-validation accuracy: 86.07%**. However, all four models achieved **83.33% on the independent test set**, so no model had a unique best test result.



Confusion Matrix

- The Decision Tree confusion matrix contains:
  - **True negatives: 3** — unsuccessful launches correctly predicted as unsuccessful.
  - **False positives: 3** — unsuccessful launches incorrectly predicted as successful.
  - **False negatives: 0** — successful launches incorrectly predicted as unsuccessful.
  - **True positives: 12** — successful launches correctly predicted as successful.
- The model correctly classified **15 of 18 test observations**, giving an accuracy of **83.33%**. It identifies all successful launches, but incorrectly classifies three unsuccessful launches as successful.

Conclusions

1. The dataset required extensive preprocessing, including extracting SpaceX launch data from the API and web pages, cleaning missing values, combining records, and creating new variables such as payload mass and landing-outcome categories.
2. Launch success varied according to launch site, orbit type, payload mass, and flight number. Certain launch sites, such as KSC LC-39A, achieved higher success rates than others in the analysed dataset.
3. Successful landings were associated with reusable booster designs, larger payload ranges, and later flight numbers. Proximity analysis also showed that launch-site facilities are strategically positioned near major transportation and coastline infrastructure.
4. Interactive visualizations helped reveal relationships that were less apparent from the tabular data. The dashboard enabled users to compare outcomes by launch site and analyse the effect of payload range on landing success.
5. Four classification models were developed: Logistic Regression, Support Vector Machine, Decision Tree, and K Nearest Neighbors.
6. The models were standardized using StandardScaler and evaluated with a **training/test split of 80/20 and 10-fold cross-validation**.
7. The Decision Tree achieved the highest mean cross-validation accuracy of **86.07%**. Therefore, the Decision Tree was selected for further evaluation as the best tuned model based on validation performance.
8. On the independent test set, all four models achieved **83.33% accuracy** using 18 observations. This shows that there is no statistically clear difference between the models on the limited test dataset.
9. The Decision Tree correctly predicted **15 of the 18 test observations**. It identified 12 successful landings and three unsuccessful landings correctly, but incorrectly classified three unsuccessful landings as successful.
10. Future improvements could include collecting more observations, testing additional algorithms such as Random Forest and Gradient Boosting, performing deeper feature engineering, and addressing class imbalance.

Appendix

- **Data Collection**
  - SpaceX REST API data collection notebook
  - Web scraping notebook
  - Extracted SpaceX launch dataset
  - API extraction flowchart
  - Web scraping flowchart

Example of API Request

```python
import requests
import pandas as pd

response = requests.get(
    "https://api.spacexdata.com/v4/launches"
)
launches = response.json()

launch_data = pd.json_normalize(launches)
```

➢ Data Wrangling

- Data-wrangling flowchart
- Missing-value replacement outputs

Missing Value Processing

```python
data["PayloadMass"].fillna(
    data["PayloadMass"].mean(),
    inplace=True
)
```

➢ SQL Exploratory Analysis

Example query for launch sites

```sql
SELECT DISTINCT Launch_Site
FROM SPACEX;
```

Example query for records beginning with CCA

```sql
SELECT *
FROM SPACEX
WHERE Launch_Site LIKE 'CCA%';
```

Example query for successful drone-ship landings with payload between 4000 and 6000kg

```sql
SELECT Booster_Version
FROM SPACEX
WHERE Landing_Outcome = 'Success (drone ship)'
  AND Payload_Mass__kg_ > 4000
  AND Payload_Mass__kg_ < 6000;
```

Example query for ranked landing outcomes

```sql
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEX
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

- **Visualization Assests**
  - Flight Number versus Launch Site scatter plot
  - Payload Mass versus Launch Site scatter plot
  - Orbit success-rate bar chart
  - Flight Number versus Orbit scatter plot
  - Payload Mass versus Orbit scatter plot
  - Yearly average success-rate line chart
  - Classification-model accuracy bar chart
  - Decision Tree confusion matrix

➢ Folium Map Assets

- Global launch-site location map
- Launch-outcome map
- Selected launch-site proximity map
- Distance measurements for railway, highway, and coastline
- Flight trajectory and proximity-line visualizations

➢ Dashboard Assets

- Launch-site dropdown
- Success-count pie chart
- Launch-site success-ratio pie chart
- Payload-range slider
- Payload versus launch-outcome scatter plot
- Booster-version colour legend

Example dashboard callback structure:

```python
@app.callback(
    Output("pie-chart", "figure"),
    Input("site-dropdown", "value")
)
def generate_launch_pie(site):
    filtered_df = spacex_df[
        spacex_df["Launch Site"] == site
    ]
    return px.pie(
        filtered_df,
        names="class",
        title=f"Launch Outcomes for {site}"
    )
```

➢ Machine-learning code

Feature preparation

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y,
    test_size=0.2,
    random_state=2
)

transform = StandardScaler()
X_train = transform.fit_transform(X_train)
X_test = transform.transform(X_test)
```

Python

Hyperparameter tuning

```python
parameters = {}
grid_search = GridSearchCV(
    model,
    parameters,
    cv=10
)
grid_search.fit(X_train, Y_train)
```

Python

Confusion matrix

```python
from sklearn.metrics import confusion_matrix

yhat = best_model.predict(X_test)
confusion_matrix(Y_test, yhat)
```

Python

Model accuracy results

Model	Cross-validation Accuracy	Test Accuracy
Logistic Regression	84.64%	83.33%
SVM	84.82%	83.33%
Decision Tree	86.07%	83.33%
KNN	84.82%	83.33%

Thank you!

